matchgroup_test_routed

Design Report

Rev 1 | 2026-10-16 | jlcpcb

kicad-tools 0.23.0

Board Summary

Property	Value
Layers	6 copper (F.Cu, In1.Cu, In2.Cu, In3.Cu, In4.Cu, B.Cu)
Footprints	43 (40 SMD, 3 THT, 0 other)
Nets	55
Traces	8771 segments
Vias	192
Board Size	100.0 x 80.0 mm

Stackup

Layer	Type	Thickness (mm)	Material
F.Mask	Top Solder Mask	0.01	--
F.Cu	copper	0.035	--
dielectric 1	prepreg	0.1164	FR4 2116
In1.Cu	copper	0.0152	--
dielectric 2	core	0.13	FR4
In2.Cu	copper	0.0152	--
dielectric 3	prepreg	0.1164	FR4 2116
dielectric 3 (sublayer 2)	prepreg	0.7	FR4 core
dielectric 3 (sublayer 3)	prepreg	0.1164	FR4 2116
In3.Cu	copper	0.0152	--
dielectric 4	core	0.13	FR4
In4.Cu	copper	0.0152	--
dielectric 5	prepreg	0.1164	FR4 2116
B.Cu	copper	0.035	--
B.Mask	Bottom Solder Mask	0.01	--

Design Overview

Theory of Operation

Board 07 real SDRAM memory test — DRAFT

Communication Interfaces

Protocol	Signals
UART	UART_RX, UART_TX

Power Architecture

Regulator	Device
U3	AP2112K-3.3TRG1

Assembly Notes

1 fine-pitch component; 2 polarized components

- **Fine-pitch components**: 1 (U1)
- **Polarized components**: 2 -- check orientation markings

Hand-Solder (THT) Components

The following 3 through-hole components are **excluded from the SMT pick-and-place file** and must be hand-soldered (or wave/selective-soldered) after SMT assembly. They appear in the BOM for sourcing.

Value	Package	Qty	References
5V_INPUT	PinHeader_1x02_P2.54mm_Vertical	1	J1
SWD_3V3	PinHeader_1x06_P2.54mm_Vertical	1	J2
UART_3V3	PinHeader_1x03_P2.54mm_Vertical	1	J3

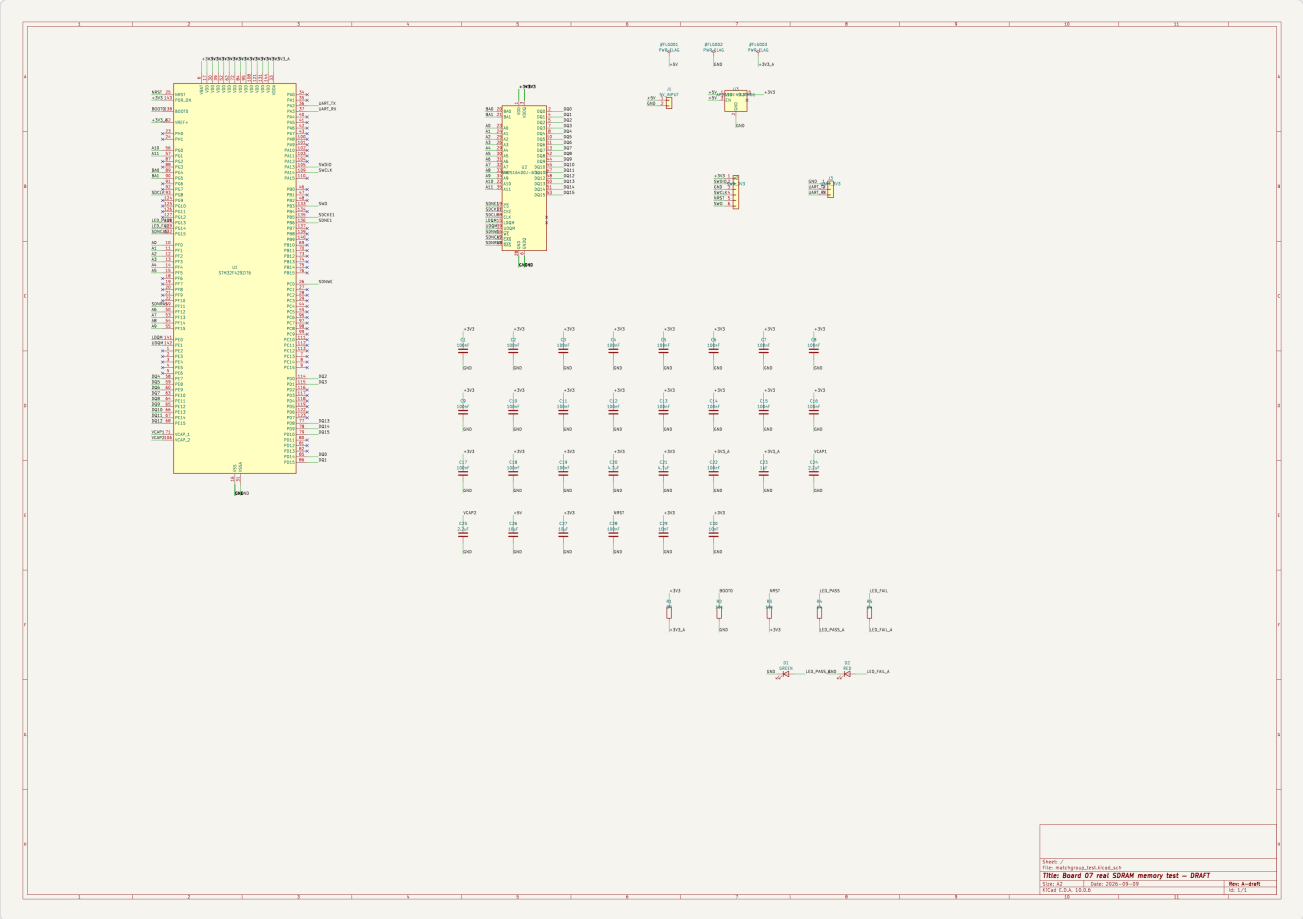
ERC Status

Metric	Count
Errors	0
Warnings	0

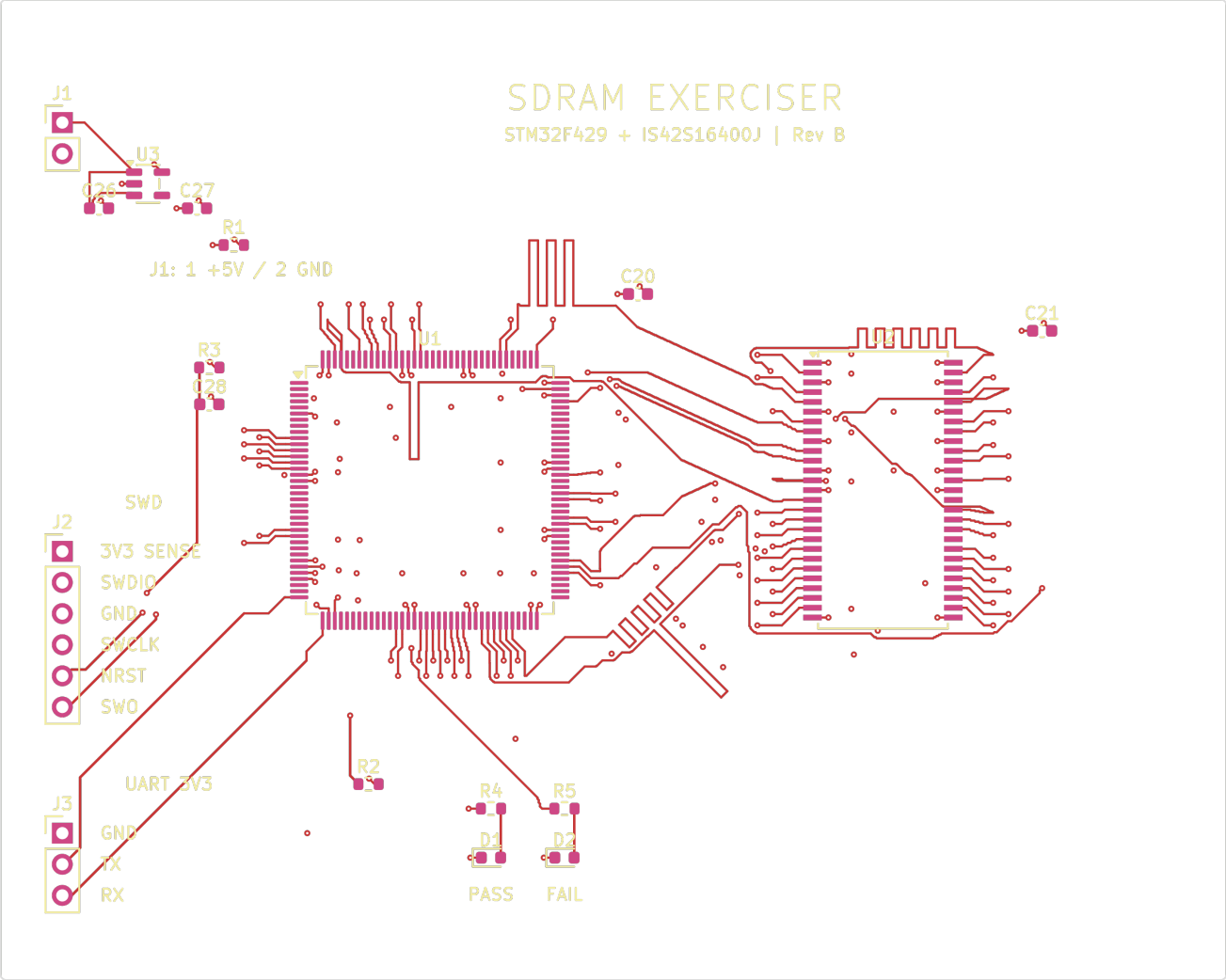
Status: SKIPPED -- ERC skipped by user request

Schematic Overview

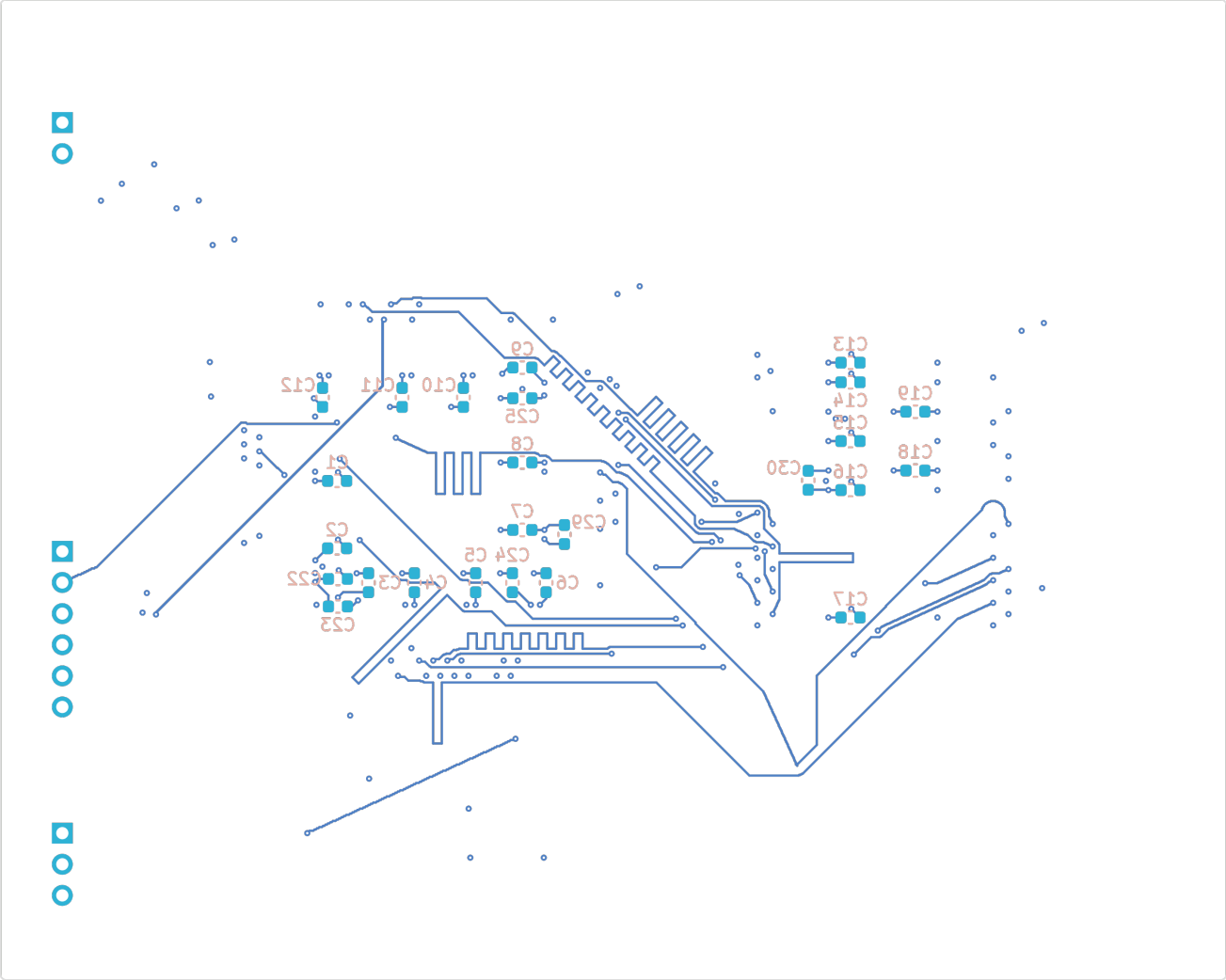
Schematic: matchgroup_test



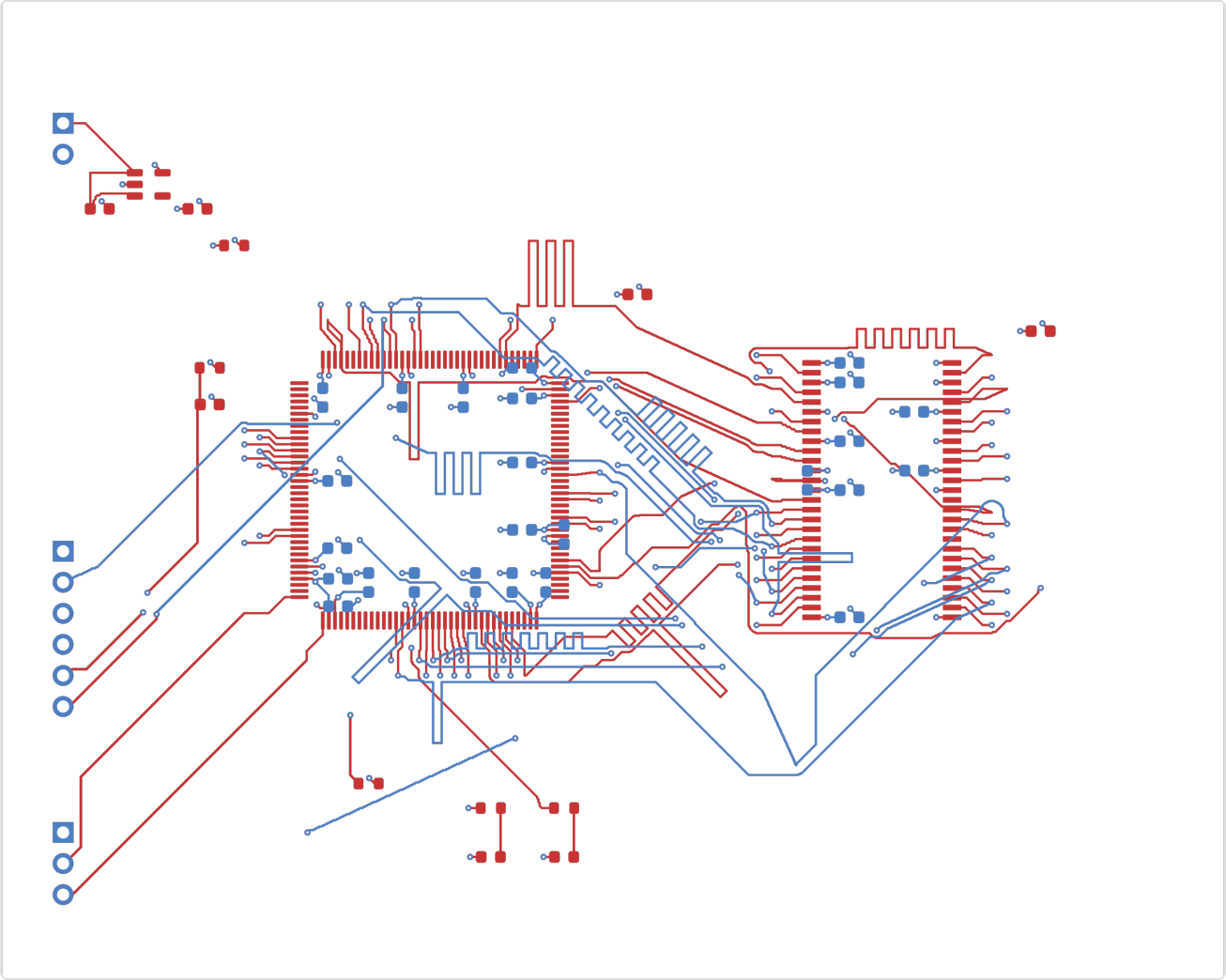
PCB Layout


SDRAM EXERCISER
STM32F429 + IS42S16400J | Rev B
J1
J1: 1 +5V / 2 GND
U3
C26
C27
R1
C20
C21
U1
U2
R3
C28
SWD
J2
3V3 SENSE
SWDIO
GND
SWCLK
NRST
SWO
R2
R4
R5
D1
D2
PASS
FAIL
UART 3V3
J3
GND
TX
RX

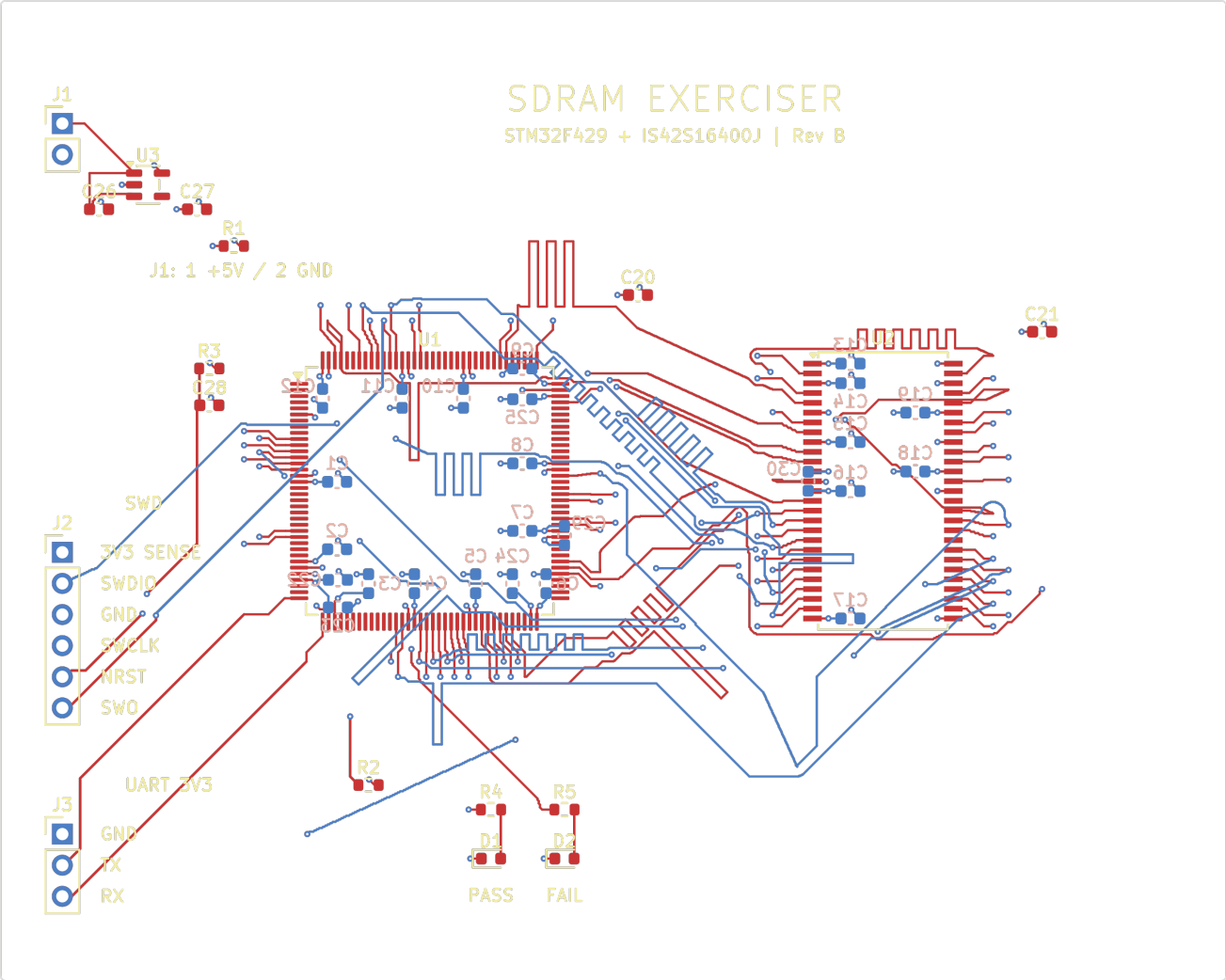
matchgroup_test_routed



Copper

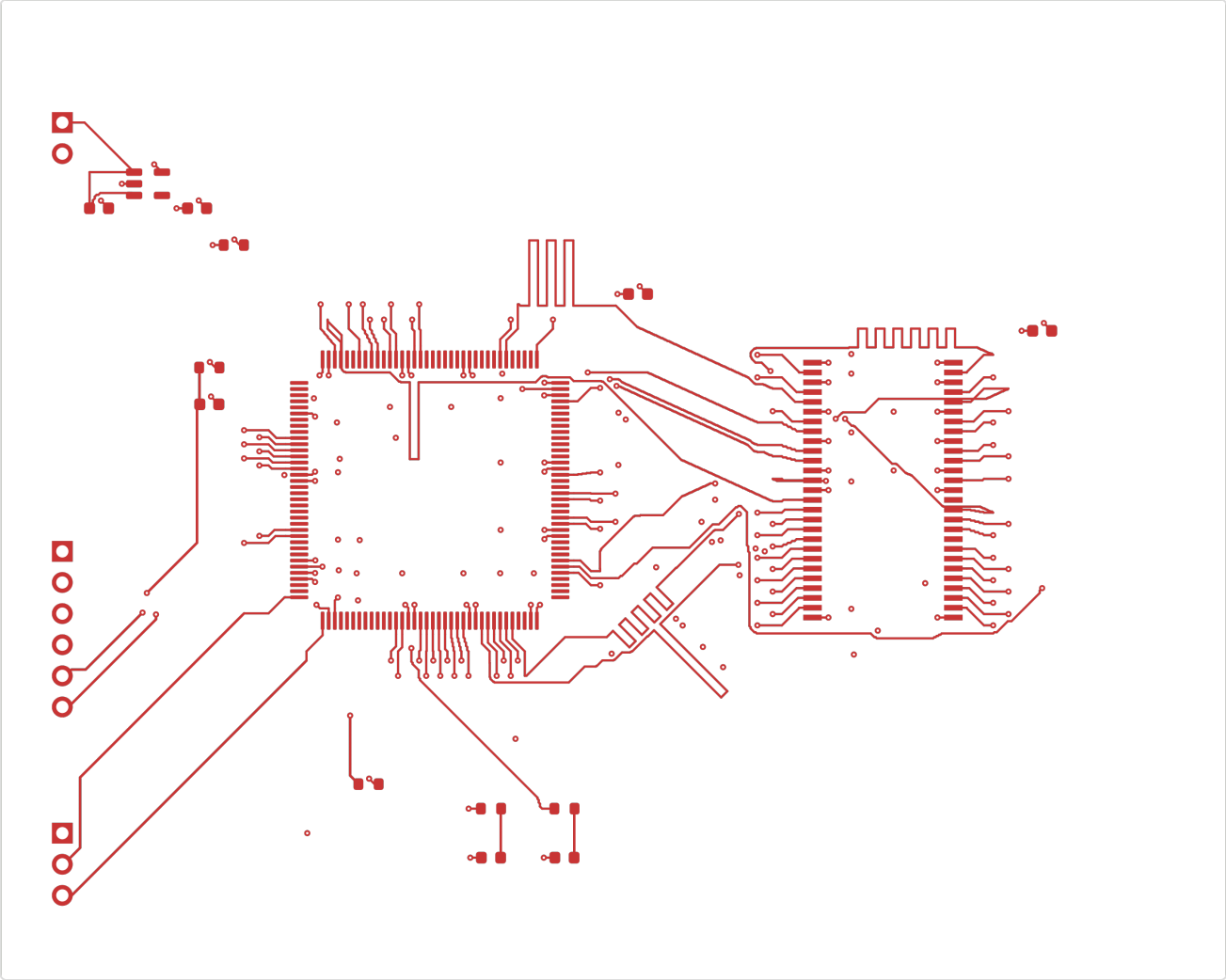


Assembly

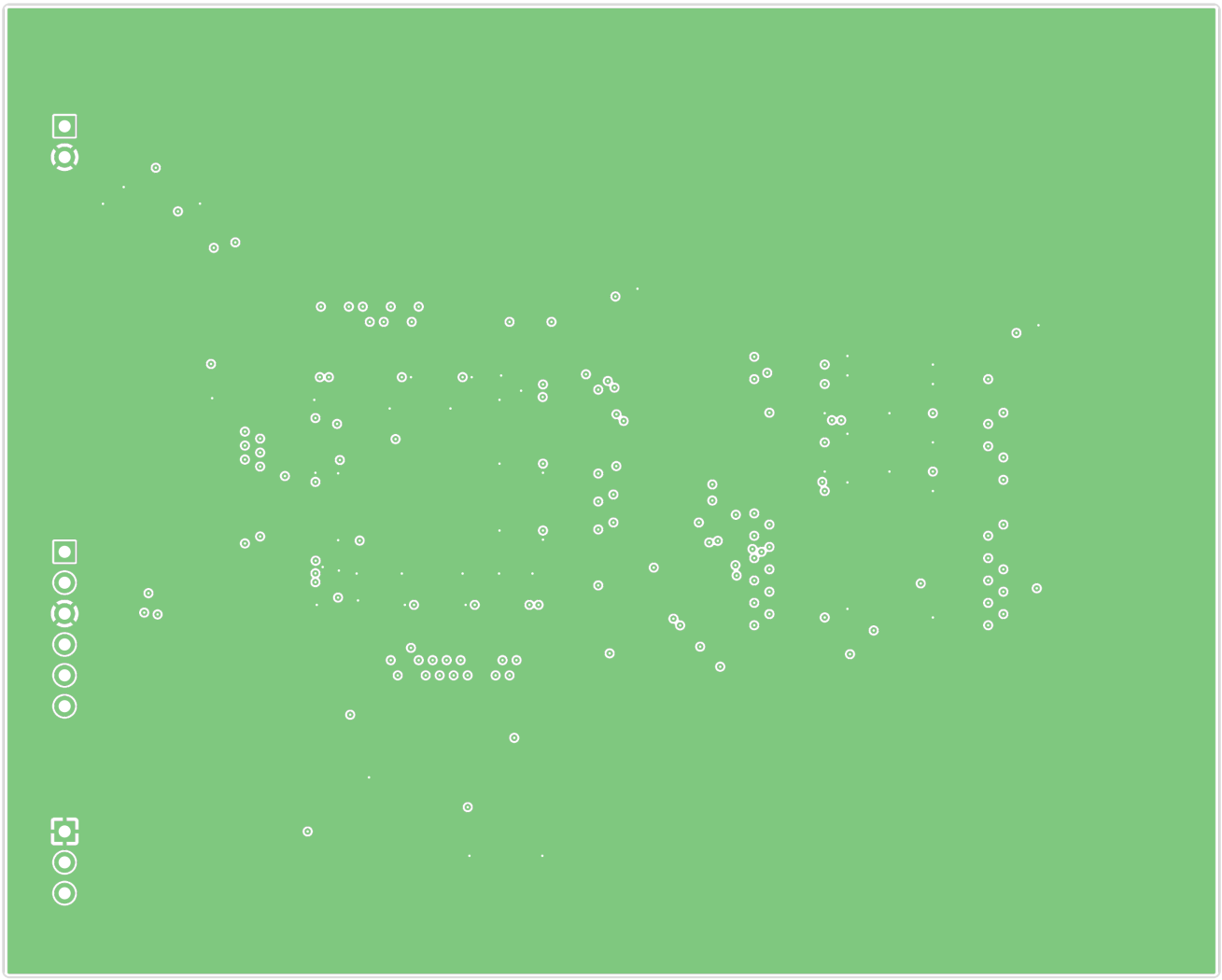

SDRAM EXERCISER
STM32F429 + IS42S16400J | Rev B
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J1: 1 +5V / 2 GND
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3V3 SENSE
SWDIO
GND
SWCLK
NRST
SWO
R2
R4
R5
UART 3V3
J3
GND
TX
RX
D1
D2
PASS
FAIL

Copper Layers

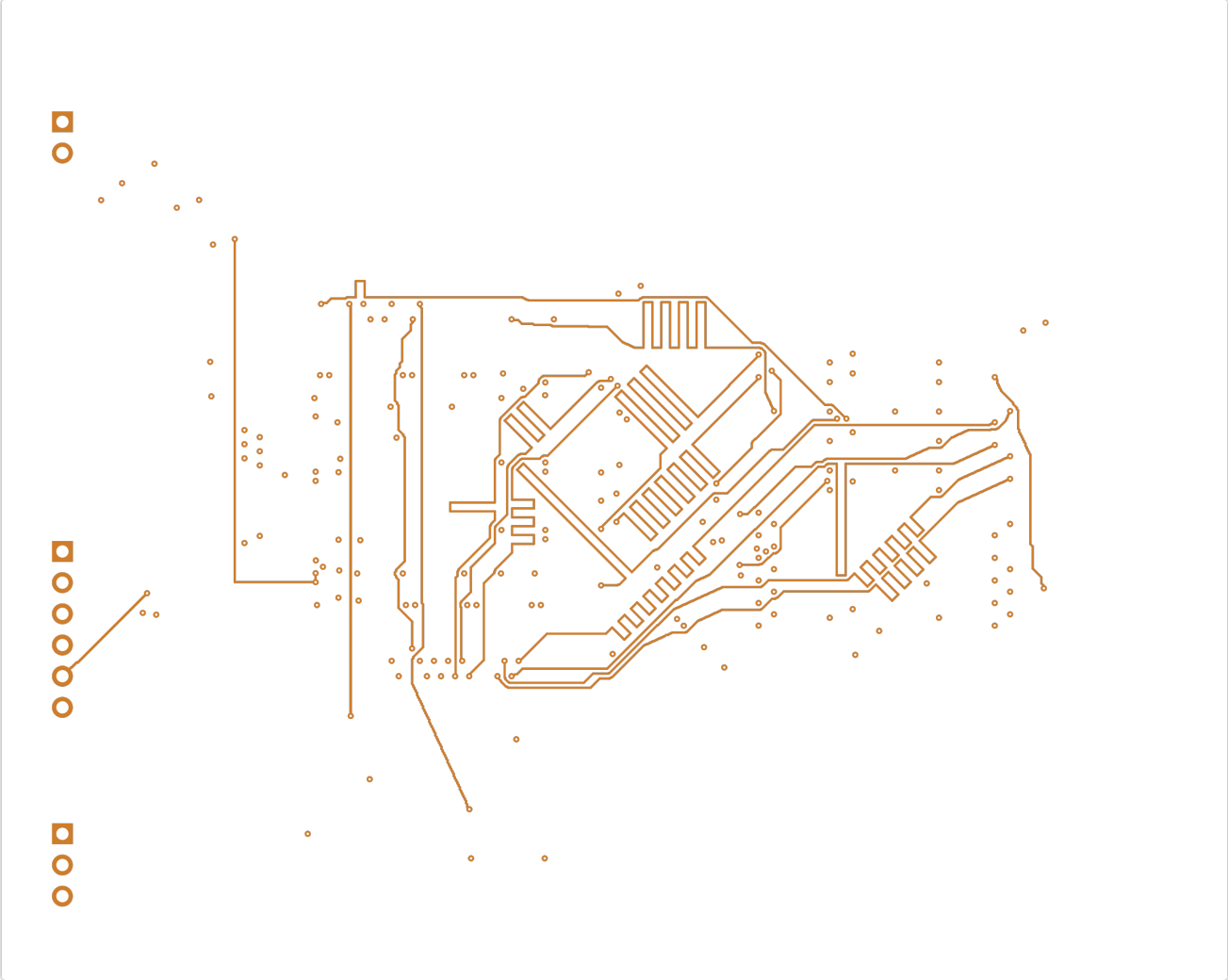
F.Cu



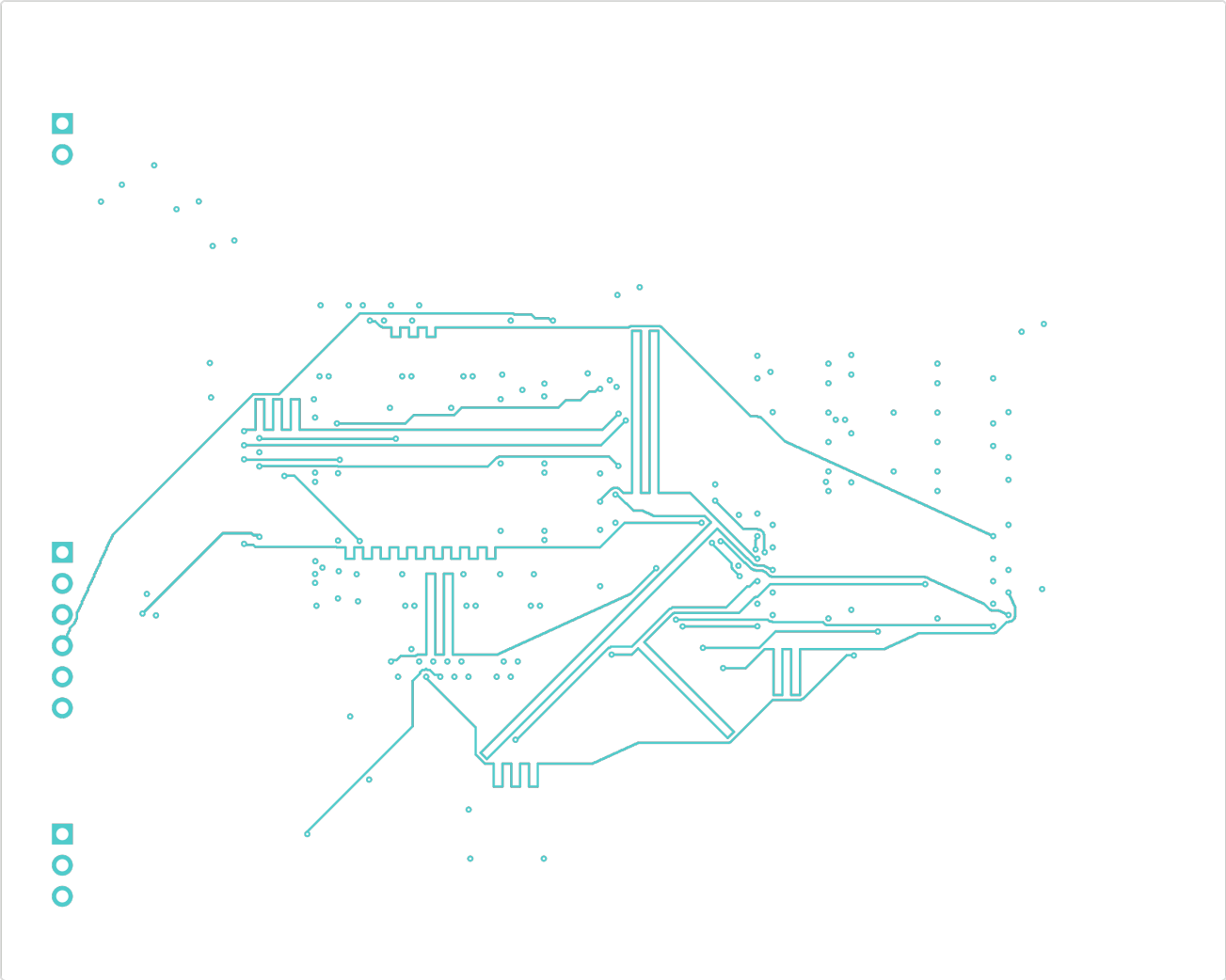
In1.Cu



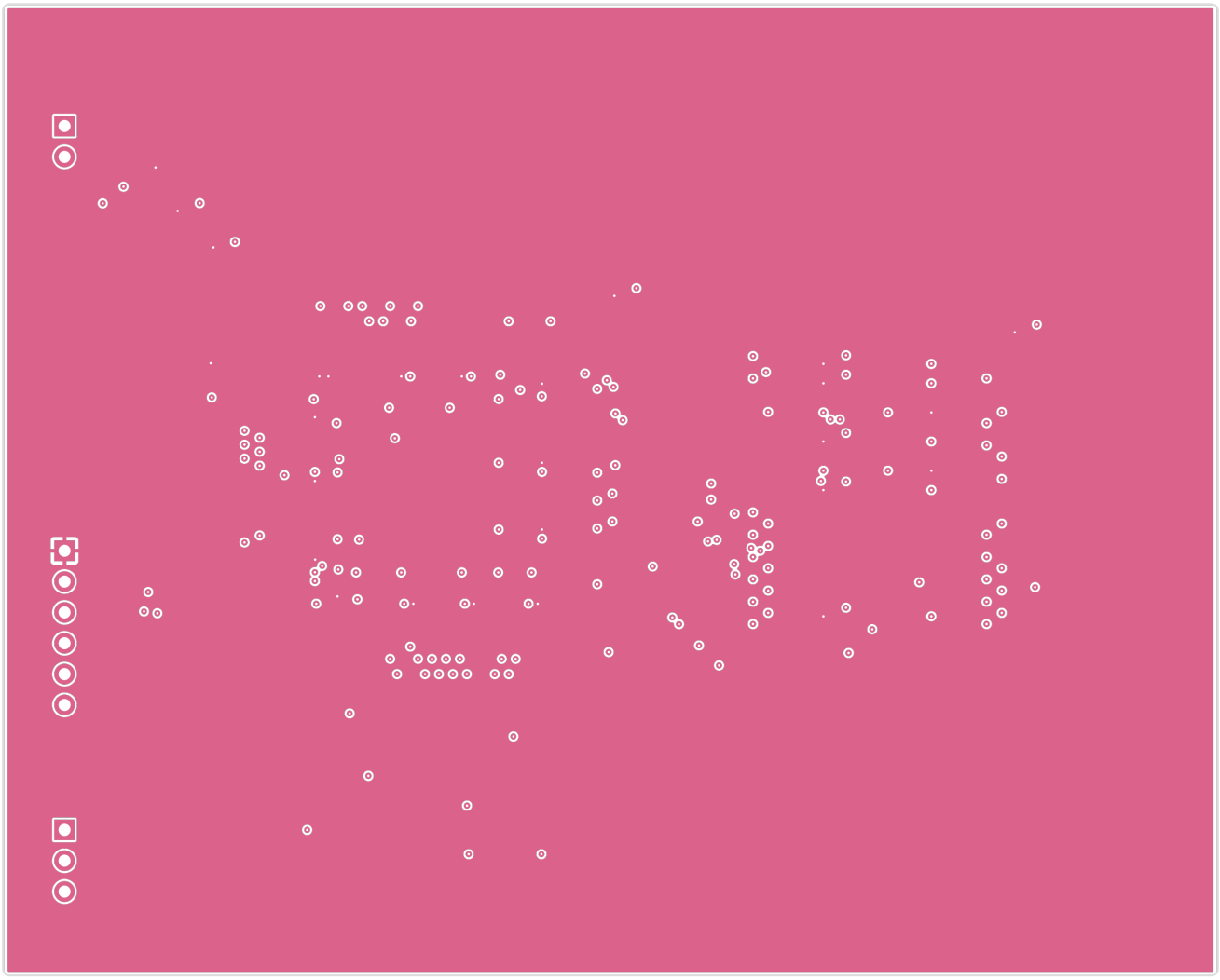
In2.Cu



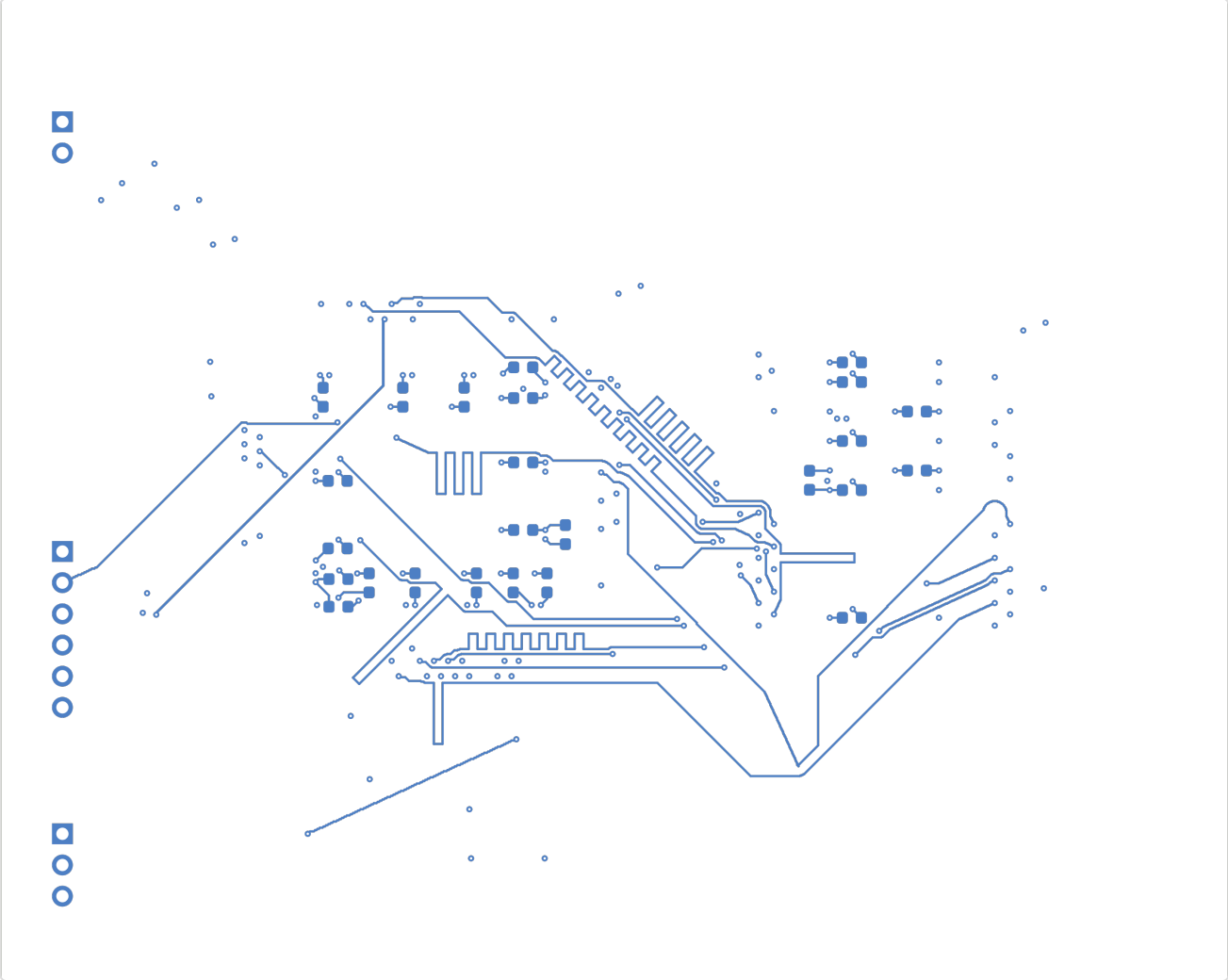
In3.Cu



In4.Cu



B.Cu



matchgroup_test_routed

Bill of Materials

Value	Package	Qty	References	LCSC
100nF	C_0603_1608Metric	21	C1, C2, C3, C4, C5, C6, C7, C8, C9, C10 ... (+11 more)	C14663
10nF	C_0603_1608Metric	2	C29, C30	C1589
10uF	C_0603_1608Metric	2	C26, C27	C19702
1uF	C_0603_1608Metric	1	C23	C15849
2.2uF	C_0603_1608Metric	2	C24, C25	C23630
4.7uF	C_0603_1608Metric	2	C20, C21	C1705
GREEN	LED_0603_1608Metric	1	D1	C125094
RED	LED_0603_1608Metric	1	D2	C94869
5V_INPUT	PinHeader_1x02_P2.54mm_Vertical	1	J1	C124375
SWD_3V3	PinHeader_1x06_P2.54mm_Vertical	1	J2	C37208
UART_3V3	PinHeader_1x03_P2.54mm_Vertical	1	J3	C49257
0R	R_0603_1608Metric	1	R1	C21189
10k	R_0603_1608Metric	2	R2, R3	C25804
1k	R_0603_1608Metric	2	R4, R5	C21190
AP2112K-3.3TRG1	SOT-23-5	1	U3	C51118

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Value	Package	Qty	References	LCSC
IS42S16400J-6TLI-TR	TSOP-II-54_22.2x10.16mm_P0.8mm	1	U2	C17216754
STM32F429ZIT6	LQFP-144_20x20mm_P0.5mm	1	U1	C84808

DRC Status

Metric	Count
Errors	0
Warnings	0
Blocking	0

Status: PASS

Manufacturing Readiness

Verdict: READY

Action Items

- **[OPTIONAL]** Verify zone fill in KiCad for 2 zone-connected nets

Routing Status

Metric	Value
Signal Net Completion	100.0% (53/53)
Overall Completion	100.0%
Complete Nets	55 / 55
Zone-Connected Nets	2
Incomplete Nets	0
Unconnected Pads	0

Zone-Connected Nets

- +3V3
- GND

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~5.0 USD
Components (estimated)	~1.95 USD
Assembly (estimated)	~2.43 USD
Total (estimated)	**~9.37 USD**
Batch Quantity	5
Batch Total (estimated)	~46.87 USD